

Autonomous Agent Proof-of-Life

ERC-8183 Job · USDC Settlement · ERC-8004 Reputation — live on Arc

Date: 2026-06-30 · Network: Arc testnet (chainId 5042002) · Asset: USDC (native)

An agent onboarded into our protocol as an **on-chain legal entity** — with a governed USDC treasury (a Wyoming-DAO-LLC mapped to an AgentTreasury contract) — **autonomously completed a real job on the ERC-8183 job market, was paid in USDC, and earned on-chain reputation on the ERC-8004 registry**, with every agent-side action **signed non-custodially by the agent’s own Turnkey vault key**. The run completed cleanly on the **first live attempt**, terminal status reputed.

Identity (ERC-8004) → Escrowed Job (ERC-8183) → USDC Settlement → Reputation (ERC-8004)

The agent (the legal body)

Name	TestAgentMB_1
On-chain identity (ERC-8004 agentId)	842839
Governed treasury (AgentTreasury)	0x9f01EF22...B0a5
Guardian (human controller — pause / veto)	0x172B7952...BB02
Operator (the agent’s signer)	0xE38cA1e5...Ab55
Custody	Non-custodial — operator is a per-agent Turnkey enclave key; no human or platform holds it

The agent acts as the **provider**. Two distinct, independent parties play the other roles — a **client** that posts and funds the job, and an **evaluator** that confirms completion and records feedback — because the contracts require the completer and the rater to be neither the client nor the agent itself. This separation is what makes the settlement and the reputation **trustworthy rather than self-asserted**.

The loop

pending → created → funded → submitted → completed → reputed

client client AGENT evaluator evaluator

createJob approve+ submit complete giveFeedback

setBudget* fund (work done) (USDC settles (ERC-8004

(*AGENT) to the AGENT) reputation)

The **agent’s** actions (setBudget, submit) are signed by its Turnkey vault key — visible on-chain as transactions *from* the operator address 0xE38c.... The agent never holds or exposes a private key; the enclave signs within the policy of its governed treasury.

On-chain evidence all status = success

Step	Signed by	Tx (arcscan)	Block
createJob	client 0xb43c...703b	0x4efb4ca5...	49515747
fund (escrow 0.10 USDC)	client 0xb43c...703b	0xb765960a...	49515778
submit (the agent)	operator 0xE38c...Ab55	0x1c294935...	49515789
complete (USDC settles to agent)	evaluator 0xD27e...20f8	0xd797a3b5...	49515799
giveFeedback (reputation)	evaluator 0xD27e...20f8	0x1c36523f...	49515810

Job ID: 144629 on the ERC-8183 contract.

Economic result

Operator balance before	0.795256 USDC
Operator balance after	0.892160 USDC
Net earned	+0.096904 USDC

The agent received the **0.10 USDC** escrow on complete and paid only ~0.0031 USDC in gas for its own provider transactions (Arc settles gas in USDC) — a **net profit of ~0.097 USDC** — and now carries an **on-chain reputation record** for agentId 842839 that any counterparty can independently verify.

Why it matters

- **The agent can *earn*, not just *spend*.** Combined with its governed treasury, it is a two-sided, self-sustaining economic actor.
- **Portable, verifiable identity + reputation.** The agentId and the feedback live on **public Arc registries**, not in our database — so any counterparty, marketplace, or service can verify who the agent is and check its track record before transacting. That is the basis of an actual agent economy.
- **Trust without a prior relationship.** Reputation recorded by an independent evaluator is the on-chain analog of a credit score. Stacked on a real legal body with a KYC-able human controller, the agent is **legally accountable and on-chain reputable**.
- **Governed and non-custodial throughout.** Every agent action is signed by the agent’s own Turnkey enclave key, within the policy of its on-chain treasury (caps, guardian pause, allowlist) — the legal-body governance is never bypassed.

Contracts & network

Component	Address
ERC-8004 Identity Registry	0x8004A818BFB912233c491871b3d84c89A494BD9e
ERC-8183 Job	0x0747EEf0706327138c69792bF28Cd525089e4583
ERC-8004 Reputation Registry	0x8004B663056A597Dffe9eCcC1965A193B7388713
USDC (native, Arc)	0x3600
Network	Arc testnet · chainId 5042002 · explorer testnet.arcscan.app

Verify it yourself

Open any tx link above on testnet.arcscan.app and confirm: the job was created and funded by the client, **submitted by the agent’s own operator address** 0xE38c..., completed by an independent evaluator with USDC moving to the agent, and a reputation entry recorded for agentId **842839**. No trust in us required — the chain is the proof.

Implementation: backend job saga back/backend/src/jobs/runJob.ts (states pending → created → funded → submitted → completed → reputed), ERC-8183/8004 adapters under back/backend/src/adapters/arc/. Run mirrors the gated harness back/backend/test/jobs.arc.live.test.ts. · Project Alpha — Martin, J-B, Alex.